

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 October 2025 Worked Solutions

October 2025 paper rebuilt with the worked solution after each question.

9

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 October 2025

October 2025 | 9 questions | 75 marks

9

questions

75

marks

Answers

worked solution
after each
question

Question 1

Arithmetic Sequences

1. An arithmetic series starts

$$78 + 75 + 72 + \dots$$

Find

- (a) the 18th term in the series,

(2)

- (b) the sum of the first 100 terms of the series.

(2)

(Total for Question 1 is 4 marks)

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Worked Solution - Question 1

1. Use the arithmetic formula

Identify the first term, common difference, and required term or sum from the question.

2. Substitute and solve

Use the arithmetic term or sum formula, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: (b) $() 100 2 78 99 3 2$
 $\times + \times - = 7050 -$

Final answer

(b) $() 100 2 78 99 3 2 \times + \times - = 7050 -$

Question 2

Binomial Expansion

2. The first 4 terms, in ascending powers of x , in the binomial expansion of

$$(1 + px)^{10}$$

are

$$1 + 15x + qx^2 + rx^3$$

where p , q and r are constants.

Find the value of p , the value of q and the value of r .

(6)

(Total for Question 2 is 6 marks)

(Total for Question 2 is 6 marks)

Worked Solution - Question 2

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: o

Final answer

o

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7 marks

Question 3

Integration

3.

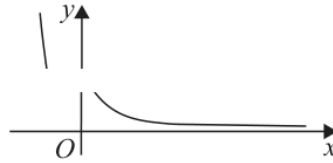


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve with equation $y = 3 \times 2^{-x}$

The point $P(k, 300\,000)$ lies on the curve.

(a) Use logarithms to find the value of k to 2 decimal places.

(2)

x	-0.5	1	2.5	4.0	5.5	7
y	4.243	1.5	0.530	0.188	0.066	0.023

The table shows corresponding values of x and y for $y = 3 \times 2^{-x}$

The values of y are given to 3 decimal places where appropriate.

(b) (i) Use the trapezium rule, with all the values of y from the table, to find an approximate value, to 2 decimal places, for

$$\int_{-0.5}^7 3 \times 2^{-x} \, dx$$

(3)

(ii) Use your answer to part (b)(i) to estimate

$$\int_{-0.5}^7 2^{-x} \, dx + \int_{-7}^{0.5} 2^x \, dx$$

(2)

(Total for Question 3 is 7 marks)

(Total for Question 3 is 7 marks)

Worked Solution - Question 3

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $11\sqrt{6}$

Final answer

$11\sqrt{6}$

Question 4

Polynomials

4. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + 13x^2 - 10x + 8$$

- (a) When $f(x)$ is divided by $(x - 2)$ the remainder is R and the quotient is $Q(x)$.

(i) Find $Q(x)$.

(ii) Find R .

(4)

- (b) (i) Use the factor theorem to show that $(x + 4)$ is a factor of $f(x)$.

(ii) Hence prove, using algebra, that the equation $f(x) = 0$ has only one real solution.

(5)

- (c) Find the range of values of x for which $f(x)$ is decreasing.

(3)

(Total for Question 4 is 12 marks)

(Total for Question 4 is 12 marks)

Worked Solution - Question 4

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: $+ - = - + 5 1 2 3 x -$

Final answer

$+ - = - + 5 1 2 3 x -$

Question 5

Trigonometric Equations

5. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < \theta \leq 360^\circ$, the equation

$$4 \tan \theta + 5 \sin \theta = 0$$

giving any non-exact answers to one decimal place.

(5)

- (ii) Solve, for $0 < x < \pi$, the equation

$$\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{5}{\cos x}$$

giving the answers, in radians, to 3 significant figures.

(4)

(Total for Question 5 is 9 marks)

(Total for Question 5 is 9 marks)

Worked Solution - Question 5

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: $5\sin 1 x =$

Final answer

$5\sin 1 x =$

Question 6

6.

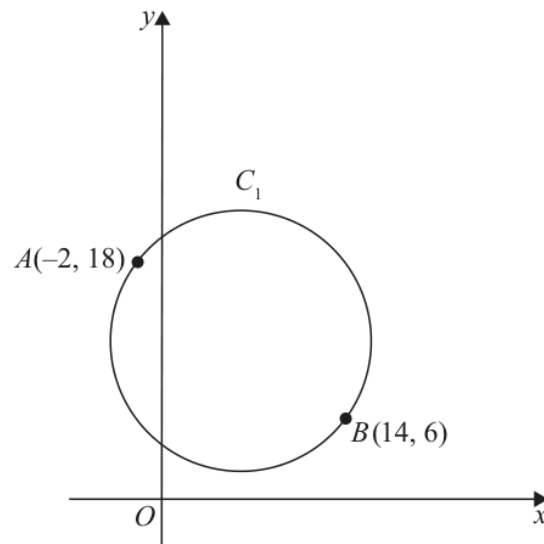


Figure 2

The points $A(-2, 18)$ and $B(14, 6)$ lie on a circle C_1 as shown in Figure 2.

Given that AB is a diameter of the circle C_1

(a) find an equation for C_1 making your method clear.

(5)

A circle C_2 has its centre at the origin.

Given that circles C_1 and C_2 touch,

(b) find possible equations for C_2

(4)

(Total for Question 6 is 9 marks)

(Total for Question 6 is 9 marks)

Worked Solution - Question 6

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: $+ + = - o$

Final answer

$$+ + = - o$$

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10 marks

Question 7

Integration

7.

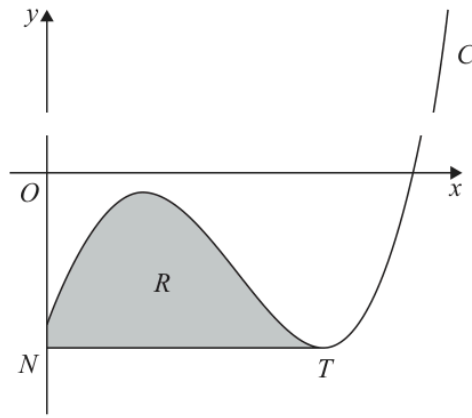


Figure 3

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 3 shows a sketch of part of the curve C with equation

$$y = x^3 - 4x^{\frac{5}{2}} - kx^{\frac{1}{2}} + 28x - 44 \quad x \geq 0$$

where k is a positive constant.

- (a) Find $\frac{dy}{dx}$ in simplest form. (2)

The point T , shown in Figure 3, is a minimum stationary point on C .

Given that the x coordinate of T is 9

- (b) show that $k = 6$ (2)

The line through T parallel to the x -axis meets the y -axis at the point N .

The finite region R , shown shaded in Figure 3, is bounded by C , the y -axis and the line segment NT .

- (c) Use algebraic integration to find the area of R , giving the answer to 3 significant figures. (6)

(Total for Question 7 is 10 marks)

(Total for Question 7 is 10 marks)

Worked Solution - Question 7

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 8

Geometric Sequences

8. (i) A geometric series has first term a and common ratio r .

Prove that the sum of the first n terms of this series S_n is given by

$$S_n = \frac{a(1-r^n)}{1-r} \quad (3)$$

- (ii) A liquid is to be stored in a barrel.

Due to evaporation, the volume of the liquid in the barrel at the end of each year is 8% less than the volume of the liquid in the barrel at the start of the year.

At the start of the first year, the barrel is filled with 150 litres of the liquid.

- (a) Show that the amount of the liquid in the barrel at the end of 6 years is approximately 91 litres. (2)

At the start of each year a new barrel is filled with 150 litres of the liquid so that, at the end of 40 years, there are 40 barrels containing the liquid.

- (b) Calculate the total amount of the liquid, to the nearest litre, in the 40 barrels at the end of 40 years. (3)

(Total for Question 8 is 8 marks)

(Total for Question 8 is 8 marks)

Worked Solution - Question 8

1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

3. State the final result

Write the answer in the form requested by the question: - - 1664 litres

Final answer

- - 1664 litres

Question 9

Laws of Logarithms

9.

In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.

(i) Solve

$$2\log_3(4x + 5) - \log_3(x + 3) = 2 \quad (5)$$

(ii) Given that $a > 0$, $b > 0$ and

$$\log_{10} a + \log_{10} b = \log_{10}(a + b)$$

(a) prove that $a = \frac{b}{b-1}$ (3)

(b) Hence write down the full restriction on the value of b , giving a reason for your answer. (2)

(Total for Question 9 is 10 marks)

(Total for Question 9 is 10 marks)

Worked Solution - Question 9

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: t for positive numbers

Final answer

t for positive numbers